

Logarithmic Notation

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With respect to a given base, **b**, **logarithms** are simply the **inverse function** to **exponents**. However, you would **never** guess that just from looking at **standard notation**. The purpose of this table is to show in a **MECE** fashion, **symmetrical** notations, and where they are **standard**, **non-standard**, or just plain **tricky**.

Exponent Notation (for base, b)	Logarithm Notation (for base, b)	Notation Combined Redundantly	Note
"the base b , raised to the ath power" " b to the ath"	"the log of a, to the base b " "log b of a"		Pronunciations
$b \uparrow a$	$b \downarrow a$	$a = b \uparrow (b \downarrow a)$ $a = b \downarrow (b \uparrow a)$	Knuth's Arrow Notation, 1976 [not ISO 80000-2] ("should" be the standard)
b^a [ISO 80000-2] (b^a in ASCII)	$\log_b a$ [ISO 80000-2] (log_b a in ASCII)	$a = b^{\log_b a}$ $a = \log_b b^a$	Common convention; Chuquet, 1484; Descartes, 1637
	$\log a$ [ISO 80000-2]		(only where b : is trivial, or already defined)
$\log^{-1} a = \log_{10}^{-1} a$ = antilog a	$\log a = \log_{10} a$ = lg a	$a = \log \log^{-1} a$ $a = \log^{-1} \log a$ $a = \log \text{ antilog } a$ $a = \text{ antilog } \log a$	[AMBIGUOUS CONVENTION] [not ISO 80000-2]
$\exp_b a$ (pow_b a in computing)	$\exp_b^{-1} a$ or $\text{antiexp}_b a$ (pow_b^{-1} a or antipow_b a in computing)	Overflowed to the Next Row	[UNCONVENTIONAL] [not ISO 80000-2]
$a = \exp_b \exp_b^{-1} a$ $a = \exp_b^{-1} \exp_b a$ $a = \exp_b \text{ antiexp}_b a$ $a = \text{ antiexp}_b \exp_b a$	$a = \exp \exp^{-1} a$ $a = \exp^{-1} \exp a$ $a = \exp \text{ antiexp } a$ $a = \text{ antiexp } \exp a$	$a = \text{ pow}_b \text{ pow}_b^{-1} a$ $a = \text{ pow}_b^{-1} \text{ pow}_b a$ $a = \text{ pow}_b \text{ antipow}_b a$ $a = \text{ antipow}_b \text{ pow}_b a$	$a = \text{ pow } \text{ pow}^{-1} a$ $a = \text{ pow}^{-1} \text{ pow } a$ $a = \text{ pow } \text{ antipow } a$ $a = \text{ antipow } \text{ pow } a$
$\exp a$ or e^a	$\ln a$	$a = \exp \ln a$ $a = \ln \exp a$ $a = e^{\ln a}$ $a = \ln e^a$	(only where b = e) [ISO 80000-2]
10^a	$\lg a$	$a = 10^{\lg a}$ $a = \lg 10^a$	(only where b = 10) [ISO 80000-2]
2^a	$\text{lb } a$	$a = 2^{\text{lb } a}$ $a = \text{lb } 2^a$	(only where b = 2) [ISO 80000-2]
2^a (only where b = 2)	$\lg a$ (only where b = 2)	$a = 2^{\lg a}$ $a = \lg 2^a$	[Chicago Manual of Style] (also some text books such as CLRS) [not ISO 80000-2]
$\exp_b^c a = (\exp_b a)^c$ (where c is a number e.g., 2 ... such that $\exp^2 a = (\exp a)^2 \neq (\exp \exp a)$)	$\log_b^c a = (\log_b a)^c$ (where c is a number e.g., 2 ... such that $\log^2 a = (\log a)^2 \neq (\log \log a)$)	Not Applicable	From trigonometry traditions [AMBIGUOUS CONVENTION] [not ISO 80000-2]